

39. Get the First 3 Rows of a DataFrame

Aim

To create a Pandas DataFrame and display the first 3 rows.

Code

```
import pandas as pd
```

```
import numpy as np
```

```
exam_data = {  
    'name': ['Anastasia', 'Dima', 'Katherine', 'James', 'Emily',  
            'Michael', 'Matthew', 'Laura', 'Kevin', 'Jonas'],  
    'score': [12.5, 9, 16.5, np.nan, 9, 20, 14.5, np.nan, 8, 19],  
    'attempts': [1, 3, 2, 3, 2, 3, 1, 1, 2, 1],  
    'qualify': ['yes', 'no', 'yes', 'no', 'no',  
              'yes', 'yes', 'no', 'no', 'yes']  
}
```

```
labels = ['a', 'b', 'c', 'd', 'e', 'f', 'g', 'h', 'i', 'j']
```

```
df = pd.DataFrame(exam_data, index=labels)
```

```
print("First 3 rows:")
```

```
print(df.head(3))
```

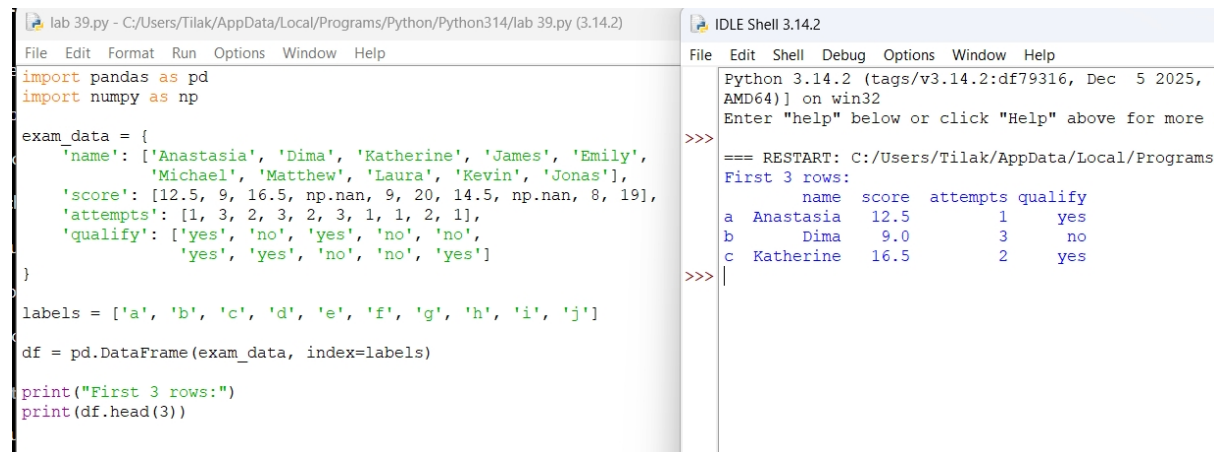
Output

```
name score attempts qualify
```

a Anastasia 12.5 1 yes

b Dima 9.0 3 no

c Katherine 16.5 2 yes



The screenshot shows a Python IDE with two windows. The left window, titled 'lab 39.py', contains the following code:

```
import pandas as pd
import numpy as np

exam_data = {
    'name': ['Anastasia', 'Dima', 'Katherine', 'James', 'Emily',
             'Michael', 'Matthew', 'Laura', 'Kevin', 'Jonas'],
    'score': [12.5, 9, 16.5, np.nan, 9, 20, 14.5, np.nan, 8, 19],
    'attempts': [1, 3, 2, 3, 2, 3, 1, 1, 2, 1],
    'qualify': ['yes', 'no', 'yes', 'no', 'no',
               'yes', 'yes', 'no', 'no', 'yes']
}

labels = ['a', 'b', 'c', 'd', 'e', 'f', 'g', 'h', 'i', 'j']

df = pd.DataFrame(exam_data, index=labels)

print("First 3 rows:")
print(df.head(3))
```

The right window, titled 'IDLE Shell 3.14.2', shows the output of the code:

```
Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025,
AMD64) on win32
Enter "help" below or click "Help" above for more
>>>
=== RESTART: C:/Users/Tilak/AppData/Local/Programs
First 3 rows:
   name  score  attempts  qualify
a  Anastasia  12.5         1     yes
b    Dima     9.0         3     no
c  Katherine  16.5         2     yes
>>>
```

40. Select the 'name' and 'score' Columns

Aim

To create a Pandas DataFrame and select only the **name** and **score** columns.

Code

```
import pandas as pd
```

```
import numpy as np
```

```
exam_data = {
```

```
    'name': ['Anastasia', 'Dima', 'Katherine', 'James', 'Emily',
```

```
            'Michael', 'Matthew', 'Laura', 'Kevin', 'Jonas'],
```

```
    'score': [12.5, 9, 16.5, np.nan, 9, 20, 14.5, np.nan, 8, 19],
```

```
    'attempts': [1, 3, 2, 3, 2, 3, 1, 1, 2, 1],
```

```
    'qualify': ['yes', 'no', 'yes', 'no', 'no',
```

```
               'yes', 'yes', 'no', 'no', 'yes']
```

```
}
```

```
labels = ['a', 'b', 'c', 'd', 'e', 'f', 'g', 'h', 'i', 'j']
```

```
df = pd.DataFrame(exam_data, index=labels)
```

```
print("Name and Score columns:")
```

```
print(df[['name', 'score']])
```

Output

	name	score
a	Anastasia	12.5
b	Dima	9.0
c	Katherine	16.5
d	James	NaN
e	Emily	9.0
f	Michael	20.0
g	Matthew	14.5
h	Laura	NaN
i	Kevin	8.0
j	Jonas	19.0

The screenshot shows a Python IDE with two windows. The left window, titled 'lab 40.py', contains the following code:

```
import pandas as pd
import numpy as np

exam_data = {
    'name': ['Anastasia', 'Dima', 'Katherine', 'James', 'Emily',
            'Michael', 'Matthew', 'Laura', 'Kevin', 'Jonas'],
    'score': [12.5, 9, 16.5, np.nan, 9, 20, 14.5, np.nan, 8, 19],
    'attempts': [1, 3, 2, 3, 2, 3, 1, 1, 2, 1],
    'qualify': ['yes', 'no', 'yes', 'no', 'no', 'no', 'yes', 'yes', 'no', 'yes']
}

labels = ['a', 'b', 'c', 'd', 'e', 'f', 'g', 'h', 'i', 'j']

df = pd.DataFrame(exam_data, index=labels)

print("Name and Score columns:")
print(df[['name', 'score']])
```

The right window, titled 'IDLE Shell 3.14.2', shows the output of the code:

```
Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:1
AMD64) on win32
Enter "help" below or click "Help" above for more info.

>>>
=== RESTART: C:/Users/Tilak/AppData/Local/Programs/Pyt
First 3 rows:
      name  score  attempts  qualify
a  Anastasia  12.5         1     yes
b      Dima    9.0         3     no
c  Katherine  16.5         2     yes

>>>
=== RESTART: C:/Users/Tilak/AppData/Local/Programs/Pyt
Name and Score columns:
      name  score
a  Anastasia  12.5
b      Dima    9.0
c  Katherine  16.5
d      James   NaN
e      Emily    9.0
f   Michael  20.0
g   Matthew  14.5
h      Laura   NaN
i      Kevin    8.0
j      Jonas  19.0

>>>
```